

Design and Implementation of a Physical Implant Attack on the Boeing 737

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Abstract

We explore a new kind of threat model for aviation cybersecurity—one in which an adversary has temporary physical access to an airframe. We argue why such attacks are practically feasible and explain how the design of existing avionics systems, their interconnects, and their maintenance architecture make such threats of particular concern. Using the Boeing 737 as an example, we develop and demonstrate a small, programmable hardware implant that can be quickly inserted into an existing maintenance socket with roughly 60 seconds of access on the ground. By carefully manipulating physical ARINC 429 bus signals, this device can create an undetected “attacker-in-the-middle” (AITM) capability between the aircraft’s flight management computer (FMC) and multipurpose control display unit (MCDU). We explore the options for addressing this class of attack and, in particular, show why transformer-coupled buses such as MIL-STD-1553, are far more challenging to manipulate in this manner.

1 Introduction

Traditional treatments of aviation cybersecurity frequently assume a fixed attack surface in which adversaries are restricted to accessing I/O channels that are explicitly public (e.g., typically wireless communications such as provided via ADS-B, ACARS, and TCAS [10, 27, 32, 33]). By contrast, attacks on individual avionics components and their hardwired inter-connections are rarely included within the threat model. Underlying this world-view is the observation that these devices are typically self-contained and hence interacting with their internal communications would require physical access—commonly deemed to be extremely low risk and hence “out of scope.” The basis for this doctrine is a combination of a strong fate-sharing argument (i.e., that if an adversary were to gain unfettered physical access to an airframe it is difficult to limit the harm they could cause) and an impracticality argument (i.e., that hangars and taxiways have strong restrictions on access and it would be difficult for an adversary to obtain the extended physical access needed to achieve such an attack).

In this paper, we *reconsider these assumptions*. In particular, we focus on the practical risks of “time limited” physical access to transport aircraft, where an attacker has a short window to gain access and install a physical implant.¹ Using the Boeing 737 as a case study, we document that the Electronic and Equipment (E&E) bay, which contains the aircraft’s key avionics components, is physically accessible to ground crew and that such a party could open the hatch, insert a custom hardware implant into an existing connector, and re-close the hatch easily within 60 seconds. Moreover, we identify at least one such connector that provides important access—notably to ARINC 429 buses that connect the pilot’s Flight Management Computer (FMC) and Multi-Function Control and Display Unit (MCDU).

Such access is not, by itself, sufficient to edit or replace existing legitimate messages on the bus—an attacker-in-the-middle capability—as simply obtaining bus access does not stop other parties from transmitting. However, as we show, the ARINC 429 bus (a workhorse for commercial and transport aviation) is electrically vulnerable to an attack where a transmitter dynamically overrides legitimate signals and covertly replaces them with signals of its choosing. We show how this can be used to synthesize a complete “virtual” attacker-in-the-middle capability without requiring splicing wires or any other form of physical mediation.²

To summarize, we make the following contributions:

- **Identifying viable physical attack surface.** By carefully reviewing 737 wiring diagrams, physical layout, and contemporary maintenance videos, we identify an Open Maintenance Connector (OMC) in the Boeing 737 E&E bay that provides access to critical ARINC 429 data buses that connect the FMC and the captain’s MCDU used to control it. Moreover, we document that this connector is potentially

¹In the US, the transport category includes jet aircraft with more than 10 passenger seats or a maximum takeoff weight of 12,500 pounds [28]. This class of aircraft comprises the vast majority of scheduled passenger service and cargo air freight service fleets.

²The core of this paper is derived from work that appeared in the 2022 Ph.D. dissertation of Sam Crow [11].

vulnerable to an attacker with short-term physical access and frame the operational requirements for such an attack.

- **Bus Driver attack.** Assuming access to the OMC, we analyze the logical and physical characteristics of ARINC 429 and identify a new attack that can achieve full “attacker-in-the-middle” capability by electrically overriding existing data signals—an approach we call a “Bus Driver” attack.
- **Empirical validation.** Based on this understanding, we build and test a prototype implant that can be quickly installed into the OMC in the Boeing 737 E&E bay. Using a testbed of authentic Boeing 737 parts, wired according to Boeing wiring schematics, we demonstrate that our implant is able to completely mediate communications between the pilot’s FMC and MCDU. This allows it to manipulate key functions such as flight plan loading and execution (i.e., where the autopilot is directed to fly) and weight and balance data (i.e., used to calculate acceptable takeoff parameters such as thrust and flap settings).

Finally, we return to the context of time-limited physical access and discuss potential countermeasures or mitigations that would prevent or further complicate such attacks in practice. In particular, we show that transformer-coupled bus designs, such as found in the MIL-STD-1553 avionics bus, are naturally resistant to Bus Driver attacks.

To be clear: while we have empirically validated that the attacks in this paper work, we do not believe they represent a threat of imminent concern (i.e., with attackers poised to quickly exploit these issues in the wild). Our goal with our research is to help alert the aviation community to this class of risks, so they may be appropriately mitigated well before they become dangerous. All of the authors of this paper routinely travel on Boeing 737 aircraft and expect to continue doing so.

The remainder of this paper is structured as follows: we first describe our overall threat model and justify its assumptions (both around physical access and technical capability), followed by an overview of the key technologies that compose our concrete attack surface in the Boeing 737 (i.e., the particular avionics components and the ARINC 429 bus). With this grounding, we then describe the theory underlying the “Bus Driver” attack in detail, along with our prototype hardware implementation. Finally, we tie these threads together by testing our hardware implementation in a 737 testbed, showing that, with OMC access, our implant is capable of fully mediating communication between key flight management (FMC) and user-interface (MCDU) components.

2 Threat Model

We assume that the attacker’s goal is to develop an implant (i.e., a device) and physically interface it with existing avionics systems such that it is able to directly or indirectly affect the integrity of flight.

Similar to Checkoway et al. [6], we consider two sets of requirements for the aviation attacker in our threat model, *operational capabilities* and *technical capabilities*. Operational capabilities reflect the challenges for an attacker to deliver a malicious implant to the target system—i.e., proximity, physical access controls, etc. In contrast, technical capabilities are the requirements on an attacker’s ability to construct a malicious implant that would have significant effect—i.e., obtaining the knowledge and documentation about target avionics systems, acquiring relevant hardware and software, analyzing and identifying points of technical vulnerability, etc. We consider each in turn, laying the ground work for the technical vulnerabilities that this paper explores more deeply.

2.1 Operational assumptions

A skilled attacker with unbounded time and access (e.g., able to modify avionics components early in the supply chain, or given extended periods of unmonitored time in an aviation maintenance hangar) will have few limitations on what they can accomplish. In this paper, we focus on a more limited (and hence higher likelihood) scenario in which the attacker’s opportunity to access critical systems without being noticed is time-limited (e.g., 60 seconds).

Here we decompose this opportunity assumption into two pieces: a) obtaining physical proximity to an aircraft, and b) obtaining access to key avionics components having gained physical proximity, and reason about their practical viability.

Airframe access. For in-service airplanes, there are three clear opportunities for physical access when a plane is on the ground and stationary at an airport: during “turnaround” between flights (approximately 30–60 minutes for short-haul flights), when parked at a gate during at the end of daily scheduled service (approximately 6 hours), and in an airport hangar as part of regular maintenance (e.g., 6–24 hours for a 737 P1 check, performed every 500 flight hours), or unscheduled maintenance that could not be completed during turnaround.³

Both the airport apron and hangars are considered sensitive areas—“Secure Identification Display Areas” (SIDA) in the parlance of the US Transportation Security Administration [29]. Airports carefully control access to these areas using physical barriers, technological controls, and human monitoring. Thus, any unauthorized adversary would need to overcome or evade these measures to approach an aircraft, let alone to do so unnoticed. That said, there is evidence that such access is feasible in practice. In a 2016 report on Airport Perimeter and Access Control, the US Government Accountability Office documented 2,500 airport perimeter and access control incidents in 2015 alone [37]. Even more concrete evidence is found in the unfortunate record of aircraft wheel-well stowaways (including multiple

³There are also extended periods of periodic maintenance, but these are typically performed at dedicated facilities and we do not consider them here.

on domestic US flights between December 2024 and December 2025) which belie the notion that existing airport security completely prevents unauthorized physical access to airframes [15, 39].

However, rather than chance being caught, a more diligent and well-resourced attacker might simply apply for a job with the appropriate credentials (i.e., an airport-specific SIDA badge). Especially during turnaround, there can be as many as a dozen different workers operating in close proximity to an aircraft; this includes ramp agents (e.g., signaling/guidance, skybridge operation, pushback, ground power, deicing, fuel, and baggage/cargo), provisioning staff (e.g., lavatory service, water service, and cleaning), caterers, and aviation mechanics. Moreover, many such jobs are both plentiful and have low barriers to entry. As of this writing, the job site Indeed.com shows 981 airport ground crew jobs available at US airports. Of these, baggage handlers, for example, typically only require a high-school diploma and a basic background check (i.e., criminal history). More skilled jobs, such as avionics technician, may require two years of training to be eligible. While these are non-negligible requirements for an attacker, we also note that a 2017 TSA report documents over 1.4 million US aviation workers with valid SIDA badges—far from an exclusive group [36].⁴

Avionics access. Proximity to an aircraft is only the first part of the operational challenge. We next consider the requirements to obtain access to a target aircraft’s avionics systems. For the purpose of this paper, we focus on a particular airframe, the Boeing 737, which is one of the most widely used transport aircraft—with over 12,000 delivered as of this writing, and almost 8,000 in service by over 459 operators today [20].⁵ In particular, our findings concern the Next Generation (NG) and MAX version of the airframe which comprise over 95% of the active fleet. *We have no reason to believe the 737 is unique with respect to the attacks in this paper; it is simply the aircraft that we have invested time, money, and effort into analyzing and testing.*

Like all modern aircraft, the 737 is a heavily computerized environment, with dozens of independent computers implicated in the flight experience. Particularly critical are the avionics systems that interface with pilots as well as

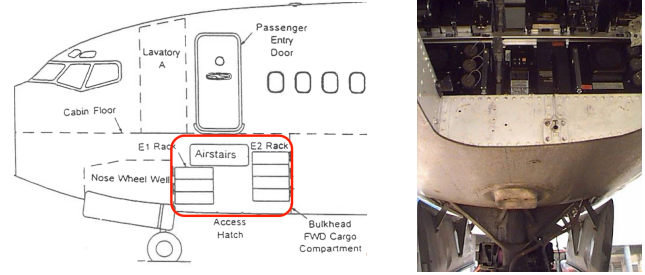


Figure 1: Boeing 737 E&E bay access hatch is located below the nose gear, within reach from the ground.

Source: Chris Brady — *The Boeing 737 Technical Site*
<http://www.b737.org.uk/>

those directly or indirectly managing control surfaces and engine thrust. The 737 is based on a traditional federated avionics architecture (also in found in the Boeing 747, 757, and 767 and in the Airbus A320 and A330 series airframes) in which a collection of individual Line Replaceable Units (LRUs) house task-specific computer functions that are then interconnected, as needed, with a combination of point-to-point wired buses (e.g., via ARINC 429) and discrete connections.

Most of these units, and much of their connectivity, are physically concentrated in one location—the Electronic and Equipment (E&E) bay.⁶ Entrance to this bay is via an external hatch designed to be easily accessible from the ground by maintenance technicians, and it is otherwise unprotected via locks or other methods of access control (Figure 1).⁷ Indeed, because of the 737’s low ground clearance, access to the E&E bay does not require a ladder or other affordance. By carefully reviewing and timing multiple videos of maintenance technicians entering the 737 E&E bay (e.g., see [3]), we empirically estimate that it should take *no more than 10–15 seconds to open the hatch and enter the bay*.⁸ Allocating 15 seconds to access the bay, 30 seconds to place the implant, and 15 seconds to close the hatch places an overall operational requirement that the attacker be unnoticed near the front of the plane’s exterior for just 60 seconds.

To summarize, the operational component of our threat model assumes that an attacker can obtain temporary (60 s) access to avionics systems. We believe that these assumptions are strong, but not unreasonable, given the range of parties who are able to gain proximity and the ease and speed with which the 737 E&E bay can be accessed from the ground.

⁴We are far from the first to make such observations. Grego writes, “Risks involving airport badgeholders grow more prevalent by the day” [18], with a similar sentiment offered by Loffi and Wallace, “Little emphasis is placed on monitoring and mitigating the security risks presented by industry employees. Once vetted, aviation employees are often granted a “trusted” status that affords them nearly unfettered access to critical aviation infrastructure, aircraft, and equipment.” [25, 38]. However, our work highlights how access risks can be further transformed by the presence of cybersecurity vulnerabilities such as those we explore here.

⁵For US airlines, the Boeing 737 comprises roughly 25% of Delta’s existing fleet, 38% of American’s, 53% of United’s and 100% of Southwest’s—and represents the plurality of all new airframe orders by each of the four airlines.

⁶This is not unique to the 737; virtually all transport aircraft have a similar concentration; e.g., the aft avionics compartment on the A320 houses most critical avionics LRUs.

⁷However, when the E&E bay hatch is opened, it activates the EQUIP light in the “door panel” section of the cockpit, but only for the duration of the time the hatch is opened.

⁸Moreover, for the purpose of the specific attacks described in this paper, we believe they can be accomplished with both feet firmly on the ground without needing to fully enter the aircraft.



Figure 2: Wiring looms in the Boeing 737. It is difficult to quickly find and cut specific wires in an unlabeled bundle of such cables.

Source: Wesley — *NextGenerationMechanic* on YouTube

<https://www.youtube.com/@NextGenerationMechanic>

2.2 Technical assumptions

We make only modest assumptions about the technical capabilities of our hypothetical attacker. In particular, we make no assumptions of special access to avionics technical materials or expertise, nor any fabrication capabilities outside those available commercially to the public. We do assume that the attacker is able to obtain appropriate software (as we have done via the secondary or grey market) and documentation (as available via online searches) to reverse engineer and understand the implementation of both standard and proprietary aviation protocols as used in the target aircraft. We assume sufficient skill to analyze wiring diagrams as well as undocumented binary data formats to identify both vectors for attack and the means to emulate existing systems. Finally, we assume a modest budget to acquire both the actual hardware used in the aircraft (for the purposes of testing and validation) and the appropriate connectors, microprocessors, and fabrication to build miniaturized implants to implement the attacks we describe. We believe these assumptions are realistic, as the prototypes described in this paper result from the efforts of a single graduate student and under \$20,000 USD to acquire the key 737 components (Section 5).

We believe that the short operational placement time requirement from our threat model inherently limits the kind of avionic interfacing that can be effective. In particular, a 30 second placement budget likely forecloses attacks that involve manipulating (tapping or splicing into) existing bus cabling. The wiring looms used in commercial aircraft contain dozens of distinct wires (e.g., see the wire loom from a 737 in Figure 2). Thus, even identifying which wires transmit messages between particular sets of LRUs is challenging—let alone splicing in a mediating implant device.

Further complicating physical manipulation, devices operating on ARINC 429 buses tend to use frequent probes to check that all bus members are operating correctly. Interrupting those heartbeat protocols leads to recorded equipment failures (e.g., a BITE error on the 737) which are stateful and flagged even if resolved. Indeed, many such buses flag individual bit errors or incorrect message spacing, such that

any splice would need to avoid creating any errors—even transient ones. For similar reasons, wholesale replacement of key avionics LRUs while the aircraft was powered would almost certainly be flagged.⁹

Towards effective avionic interfacing, we note that there are a variety of existing avionics data connectors—both directly on individual LRUs and broken out separately—that are provided for maintenance purposes. Such connectors provide an independent path to accessing key buses and are both easily identifiable and accessible. Thus, we assume that attackers will focus on misusing such connectors and how they might be manipulated via an implant to achieve their goals.

2.3 Comparison with physical attacks

It has been suggested that an attacker with physical access, such as described in this paper, could simply mount physical attacks (e.g., planting explosives). This is decidedly true, and is a risk that deserves independent attention. However, the threat posed by an implant is distinct in both capability and flexibility. First, an implant-based attack supports non-destructive threat scenarios. For example, an attacker might cause a plane to deviate from its flight plan and fly into another country’s airspace, thus enabling that country to detain a passenger.¹⁰ Second, even when destruction is the explicit goal, an implant can provide a measure of deniability. For example, if a plane crashed on takeoff due to bad weight and balance settings or was lost over the ocean due to a lack of fuel, this might be reasonably blamed on pilot error or equipment faults. Third, an implant also enables remote interactive control for scenarios in which timing or flexibility is important. Finally, there are a range of measures used for detecting explosives in the airport environment, but these would be ineffective at interdicting an implant such as the one we describe here.

3 The 737 OMC Attack Surface

As mentioned earlier, we have identified an Open Maintenance Connector (OMC), located in the E&E bay of Boeing 737 (NG and MAX) airframes, that provides direct access to key buses for the airplane’s Flight Management Computer (FMC) and Multi-Function Control and Display Unit (MCDU).¹¹ In this section, we describe the function of these

⁹In addition, although LRUs are racked in trays with rear-mounted connectors (typically similar in design to ARINC 600), so they may be replaced in the field, they are heavy (e.g., the FMC for the 737 weighs 25 pounds), and they are secured by multiple restraining screws. Reliably replacing an LRU in 30 seconds would be challenging.

¹⁰Ryanair Flight 4978 offers an illustrative example of this scenario, in which a passenger plane transiting Belarusian airspace en route to Lithuania was diverted to the Minsk airport under false pretenses—a bomb threat—where authorities then arrested Belarusian journalist Roman Pratasevich and his partner Sofia Sapega [22].

¹¹Note: OMC is a generic acronym we invented to obscure the precise identity and location of the vulnerable access connector.

units and the nature of the attack surface exposed by the OMC (i.e., and therefore available to an attacker’s implant with time-limited access to the E&E bay). In the subsequent two sections we will amplify this raw capability, first, by manipulating the ARINC 429 bus protocol to synthesize complete attacker-in-the-middle functionality and, second, by constructing a complete attack by carefully reverse engineering the proprietary protocols used to communicate between the FMC and MCDU.

3.1 Flight Management Computer

The standard General Electric FMC used on the 737 calculates flight plans between airports that pass through predefined waypoints and that follow standard departure and arrival paths. In flight, it commands the autopilot and autothrottle to follow the planned route. The FMC also calculates important performance numbers, like the minimum runway length the plane needs to safely take off, the “V speeds” for takeoff that determine critical go/no-go decisions and climb performance, and the slowest speed the aircraft can fly without stalling. Since it is critical for safe and efficient flight, there are two, redundant FMCs. The default is the captain’s FMC, and if that fails a switch can transfer control to the first officer’s FMC. There is a separate transfer bus that transparently keeps the two FMCs in sync (i.e., such that updating the state of one FMC will cause the other to reflect the update as well).

3.2 Multi-function Control and Display Unit

The MCDU, also made by General Electric, is the standard terminal that interfaces between the pilots and a range of other LRUs, including the FMC. It has an integral alphanumeric keypad and screen that the pilots use to enter and view a wide array of information including, for our purposes, the flight plan, and weight and balance data. The FMC is typically structured around individual “pages” that focus on particular pieces of information, with buttons alongside the screen used to navigate through the page hierarchy. As with the FMC, there are two MCDUs on the flight deck, one for the captain and one for the first officer. Importantly, each MCDU has a dedicated ARINC 429 bus that it uses to send button press events to the FMC, and another bus where it receives text from the FMC to show on its screen. The two MCDUs operate independently and can show different information.

3.3 Open Maintenance Connector

To support maintenance tasks, the buses connecting the captain’s FMC and MCDU branch off to the OMC inside the E&E bay via yet another LRU (the Integrated Flight Systems Accessory Unit, or IFSAU). As described in our threat model, anyone on the ground around the plane can reach the OMC through the unprotected E&E bay hatch.

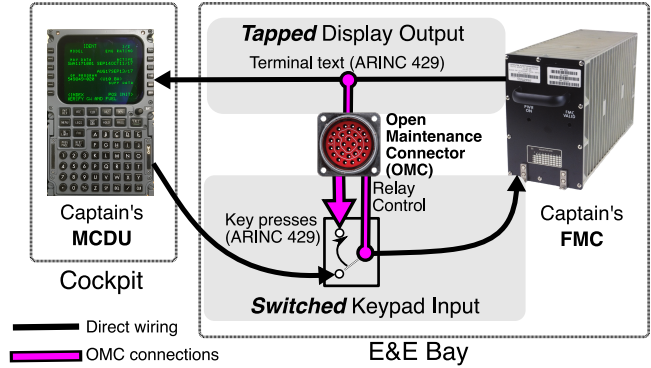


Figure 3: An Open Maintenance Connector (OMC) located in the E&E bay of the 737 can take complete control over keystrokes sent to the sensitive Flight Management Computer (FMC). It is tapped into the bus that outputs text to the MCDU (aka cockpit display), so it can inject additional signals into the bus while the FMC is *also* updating the screen.

Figure 3 shows how the OMC exposes a key attack surface in the 737. It is connected in the middle of *two* ARINC 429 data buses: (1) it is *tapped* into the bus from the FMC to the MCDU, to reveal the text that the FMC sends to the MCDU display, and (2) it is *switched* into the bus from the MCDU back to the FMC—by means of a relay controlled by a discrete signal in the OMC—to allow sending key presses. When the relay is activated, the FMC receives key presses from the OMC instead of from the captain’s MCDU. Notably, none of this functionality is predicated on being on the ground, and thus an implant inserted into the OMC will have the same capability during flight.¹²

3.4 Abstract Attack

The attacker’s goal is to insert an implant into the OMC that allows them to covertly inject messages (e.g., key presses) to modify the state of the FMC, while hiding such updates from the captain’s view by preventing FMC state changes from appearing on the MCDU. Part of this task is straightforward. The relay in OMC’s design allows an implant to silently and transparently inject button presses from the MCDU to the FMC (i.e., changing flight waypoints, weight and balance, etc.). However, unless the captain could be reliably distracted, such an attack would be quite obvious, since each input (and any output it subsequently caused) would be reflected on the captain’s MCDU—like a ghost pressing buttons. To deliver an attack covertly requires *also* controlling the messages the MCDU receives from the FMC. However, this step is consid-

¹²We first disclosed these issues to Boeing in the first half of 2020. Since then, we have provided them with multiple detailed presentations as well as making our team and test hardware available for testing and validation in their 737 testing facilities. We have maintained a close and continuing discussion with them about the issue and they have reviewed a near-final draft of this paper. Further context about disclosure can be found in Appendix C.

erably more complex because there is no relay isolating the MCDU from the true FMC’s messages.

The next section addresses this limitation and provides both the theory and an empirical demonstration of a “Bus Driver” attack that can intercept and alter the display bus entirely through the tapped access the OMC provides. We show that the electrical design of the ARINC 429 aviation bus is vulnerable to a new class of attack that allows any party on the bus to synthesize a complete attacker-in-the-middle capability. Indeed, due to this capability, the presence of the relay (that effectively substitutes in the OMC on the MCDU→FMC path) is a mere nicety; our “Bus Driver” attack could synthesize equivalent behavior if the bus was shared (instead of switched).

4 The Bus Driver Attack

Typically, an attacker-in-the-middle (AITM) capability requires physical mediation of a channel—being able to both intercept unwanted messages and inject new messages in their place. However, in this section we will describe an alternative approach, which we call a “Bus Driver” attack. This attack allows *any* party on the bus, whether or not they are “in the middle” physically, to synthesize a complete AITM capability by exploiting assumptions in the bus’s electrical design. In the context of this paper, our Bus Driver attack exploits design assumptions in the ARINC 429 bus [2] in order to mediate messages sent from the FMC to the MCDU (i.e., and to ultimately to control what the pilot sees on the screen).

4.1 ARINC 429 background

ARINC 429 is a serial bus invented in 1977 for safety-critical applications that is widely used in commercial and transport aircraft (including the 737). Unlike many common industrial buses, such as CAN, ARINC 429 is a single-master bus (i.e., there is a single transmitter and a small number of receivers). For this reason, a message override attack on ARINC 429 was previously thought to be highly challenging, as it was assumed that an attacker would need to unplug a legitimate 429 transmitter and replace it with a malicious one [17]

The ARINC 429 protocol is unidirectional. Some single-master bus designs, such as MicroLAN (1-Wire), feature protocol-level controls where the master designates ‘which way’ data is flowing (i.e., which device is transmitting). In ARINC 429, if a pair of devices require two-way communication they must use two buses, one in each direction.¹³ Data words on ARINC 429 are 32 bits in length. Due to the legacy nature of the protocol, there is no message authentication or other integral data security features.

¹³Indeed, in traditional “federated avionics” designs (Boeing aircraft before the 777, and Airbus aircraft before the A350) there is no common communication medium between LRUs and they are interconnected with as many independent 429 buses as needed.

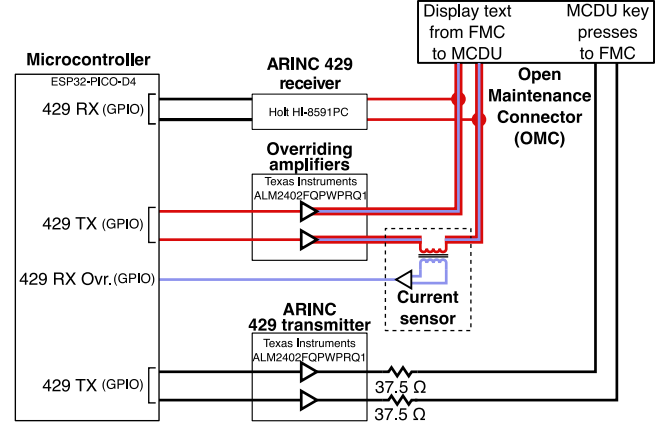


Figure 4: A block diagram of the 429 Bus Driver implant. When the Bus Driver is active, its output amplifiers drive current through the bus. A current sensor measures that current which, in turn, allows the Bus Driver to infer the contents of any other signal being sent simultaneously.

At the physical layer, ARINC 429 is a 100 kbps differential bus carried over a 75 Ω twisted pair cable.¹⁴ Transmitters connect to the two bus wires through a set of 37.5 Ω resistors [2]. These resistors serve two purposes: (1) providing an output impedance that reliably matches the (approximately 75 Ω) characteristic impedance of the twisted pair bus wires; and (2) protecting the transmitter during a fault (e.g., short circuit) by limiting the current that can flow into or out of the transmitter to a safe level.

4.2 Basic vulnerability

Even though 429 is designed for a single “master”, the transmitter design does not *prevent* a malicious device from transmitting as well (e.g., if a hardware implant gained physical bus access via an open connector). Moreover, as we will describe, the impedance-matching resistors on the legitimate transmitter, designed to improve reliability, ultimately create the opportunity for such a malicious transmitter to “override” any existing legitimate signal.

The general principle of the Bus Driver attack is that a malicious transmitter drives more current on the bus than the legitimate transmitter is able to, thus overriding the legitimate signal. In addition, by measuring how much current it is driving, the malicious transmitter can *also* decode the legitimate transmissions actively being overridden.

Figure 4 shows the major components of an ARINC 429 Bus Driver implant. At its core, the Bus Driver is simply an ARINC 429 transceiver driven by a microcontroller (MCU). The transmitter is augmented with two high-current amplifiers (i.e., drivers) and is connected directly to the bus *without* the addition of the impedance matching (37.5 Ω) resistors.

¹⁴There is also a 12.5 kbps mode, but it is not used in our application.

With this lower-impedance connection to the bus and reasonably strong amplifiers, the Bus Driver can “override” the transmissions on the wire from the legitimate transmitter.

To achieve full AITM capability the Bus Driver also receives legitimate transmissions simultaneously while it is overriding. To do this it includes a current sensor and comparator, attached to the transmission bus after the amplifiers. This allows the Bus Driver to read how much current the overriding amplifiers are driving and, from that, the MCU can decode what bits were being overridden.

4.3 Underlying Theory

Thus far, we have been informal with the word “override” and here we explain in more detail how electrical current flows across an ARINC 429 bus during the Bus Driver attack.

Normal Operation. ARINC 429 is essentially an unterminated differential bus. The role of the inline $37.5\ \Omega$ resistors at the transmitter is to absorb the first reflection from the unterminated end of the bus. The current flow through an unterminated bus is also dictated by the impedance of the transmission line, which is a twisted pair $75\ \Omega$ cable in ARINC 429. When an ARINC 429 transmitter is actively transmitting on a bus it applies $+5\ \text{V}$ or $-5\ \text{V}$ across both drivers. The signal encounters impedance immediately because of the $75\ \Omega$ impedance of the twisted pair cable (until the wave propagates to the other end of the cable) and the inline resistors. This means that the maximum current a normal ARINC 429 driver and its resistors encounter is only 89 mA,

$$I_{\max} = \frac{V_{\text{step}}}{R_b + Z_0} = \frac{10\ \text{V}}{37.5\ \Omega + 75\ \Omega} \approx 89\ \text{mA}.$$

Note, the drivers and resistors are not stressed in normal operation because the current flows only briefly due to the short duration of bit transmissions ($10\ \mu\text{s}$ per bit at 100 kbps).

Override attack. The $37.5\ \Omega$ inline resistors are what make this bus vulnerable to a Bus Driver implant. They essentially serve to limit and absorb the current that the legitimate transmitter supplies to the bus, allowing the more powerful Bus Driver to control the voltage on the bus and therefore dictate what bits are transmitted. We describe the current flow in the most extreme scenario for an override attack: when opposite bits are sent by the legitimate transmitter and the Bus Driver. Assuming the transmissions are synchronized, the legitimate transmitter will attempt to put $+5\ \text{V}$ on one wire through its inline $37.5\ \Omega$ resistor, and the Bus Driver at the tapped point will put $-5\ \text{V}$ on the same wire (or vice versa). Both transmitters will briefly encounter the characteristic impedance of the wire, but instead of the current peaking at that point, the current will peak later when the wave fronts reach both transmitters, where the current across the bus will spike to

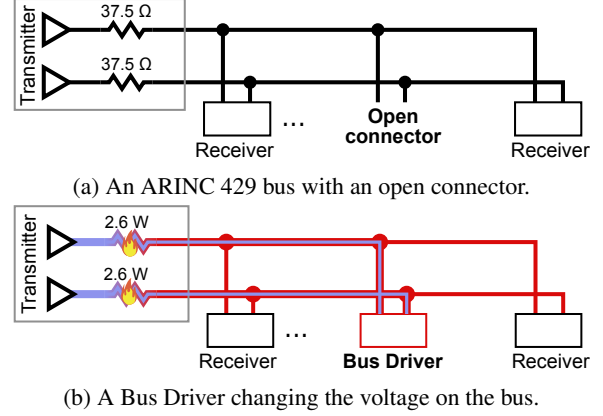


Figure 5: Comparison of normal and implant-overriding operation. The thick lines represent the current that flows between the implant and transmitter. The resistors in the transmitter limit the current and protect the transmitter from overheating.

267 mA per wire for the duration of the bit ($10\ \mu\text{s}$),

$$I_{\text{fight}} = \frac{10\ \text{V}}{37.5\ \Omega} \approx 267\ \text{mA}.$$

Without any resistors, both transmitters would push the bus toward infinite current, likely burning up one or both transmitters. The resistors prevent the legitimate transmitter from sinking too much current, which makes it reliable and protects it from damage, but also effectively blocks the signal from the legitimate transmitter. Current from Bus Driver causes each resistor to dissipate up to 2.67 W of heat (Figure 5b). Although this is likely well beyond their designed heat output, they only have to do this for the microsecond-scale duration of each bit and they will have time to cool when both transmitters are trying to transmit the same bit or when the transmissions end. In our tests, these resistors were able to dissipate this power without issue.

By measuring the current draw on one of its 429 bus wires, the Bus Driver can *also* support a full AITM by inferring what bits the legitimate transmitter attempted to send, based on whether the implant is actually overriding or coincidentally sending the same value. If there is minimal sustained current draw, then both are transmitting the same bit. If there is non-negligible sustained current draw,¹⁵ the legitimate transmitter is transmitting the opposite bit. Note that this concept extends both to actively overriding a transmission, or simply suppressing one; Figure 6 shows an example of decoding what the legitimate transmitter tried to send when the implant is holding the bus idle (i.e., in the NULL condition, with nominally 0 V difference across the bus wires).

¹⁵Note: In practice, override current will be less than 267 mA; it is bounded by the drive strength of the transmitters and various non-idealities. These factors are also why there is nearly a $2\times$ difference in current required to force a HI versus LO condition to NULL in Figure 6.

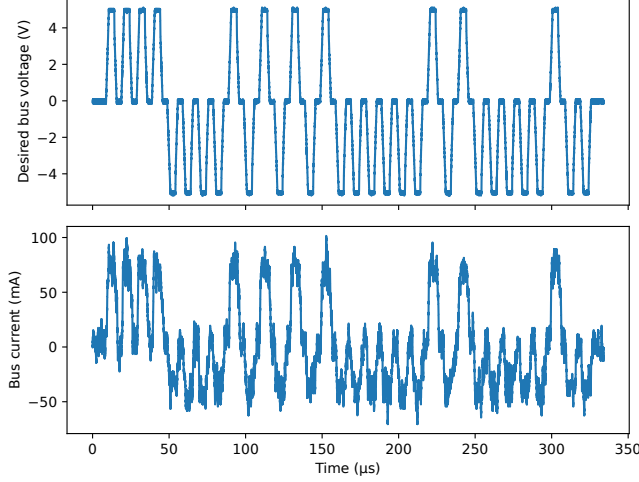


Figure 6: When the FMC tries to send a message (top) that the Bus Driver blocks, the current readings (bottom) from an Allegro ACS70331 current sensor allow the Bus Driver to determine what message the FMC is trying to send.

4.4 Implementation

We designed and built a proof-of-concept ARINC 429 Bus Driver which can allow, block, or modify legitimate ARINC 429 transmissions. In addition, our implant can read the messages that the legitimate transmitter sends and at the same time prevent any other receiver on the bus from receiving a message. The Bus Driver is driven by an ESP32 microcontroller which operates as a bit-banged ARINC 429 transmitter.

The first challenge to overcome was synchronizing the Bus Driver with the legitimate transmitter. This is necessary so the implant can transparently AITM the channel on a per-bit basis; receivers must not observe any messages with errors in them because only part of the bits were overridden. To this end, we constructed a feedback loop between the bit banded overriding transmitter and receivers for legitimate transmissions in firmware on the microcontroller. The firmware monitors the receiver to see when the legitimate transmitter is sending bits and the timing is synced with the overriding transmitter using the current-sense receiver. Since ARINC 429 messages are structured with a message type before the payload, we can use this to sync during the transmission of the message type and override the payload with the attack payload.

We found we could achieve tight enough sync in the ESP32’s firmware to send and receive at 100 kbps using the following configuration. We use the MCU’s motor control pulse width modulation peripheral to measure the duration of each bit in the message and trigger an interrupt when an input changes. The interrupt handler checks that each bit has the correct timing and collects the 32 bits of each message into one integer value for processing. This works reliably as long as the microcontroller is running at its maximum frequency of 240 MHz. Otherwise, the interrupt handler takes too long

and we miss some messages. To transmit, we use bit banging. Because the MCU’s software libraries do not provide a way to wait with a resolution finer than 1 μ s, and to avoid timing errors caused by interrupts, we do not use either of the MCU’s main cores. Instead, we use the ESP32’s Ultra-Low-Power (ULP) coprocessor. The ULP coprocessor has a restricted instruction set that makes it challenging to program, but it can wait for a precise number of clock cycles and never gets interrupted by other software. The ULP software controls two digital outputs, which the output amplifiers convert into positive and negative voltage on the ARINC 429 bus.

4.5 Limitations

While a Bus Driver attack is able to effectively control which symbols an ARINC receiver decodes, it is not *perfectly* stealthy. In particular, since the attack works by overriding the legitimate transmitter, the waveform produced by combining the legitimate transmitter’s signal with the Bus Driver’s will have slightly different *electrical* characteristics when compared to the legitimate transmitter’s signal alone. In Section 6, we describe several approaches to detecting such an overridden signal, including physical-layer voltage-based detection systems of the sort described by G  n  reux et al. [19].

5 Empirical attack on the 737

We have explained that the 737’s Open Maintenance Connector (OMC) can provide an attacker with both control over the inputs to the FMC (via a relay) and the outputs to the screen (via the Bus Driver attack). However, this is still somewhat abstract. There yet could be details in the 737 FMC, MCDU or IFSAU hardware or software that preclude such an attack or safety mechanisms that would alert to its presence.

Thus, in this section, we implement and empirically evaluate a complete end-to-end attack using the actual hardware employed in the 737, wired according to Boeing’s schematics. After describing our testbed, we then describe the interactions between the FMC to MCDU display protocol and the challenges of its interaction with the Bus Driver attack. Following this context, we explain our end-to-end attack implementation, which can both selectively inject malicious button presses to the captain’s FMC and alter text as needed (i.e., concealing state changes) on the captain’s MCDU. We then describe our efforts in making a practically deployable implant (via miniaturization). Finally, we explore the ramifications of the implant’s capability and how it can be used to impact several safety-critical aspects of flight.

5.1 Testbed

We evaluated the attack on the genuine avionics used in contemporary models of the 737 (i.e., NG and MAX). We purchased genuine 737 FMC and MCDU avionics from parts

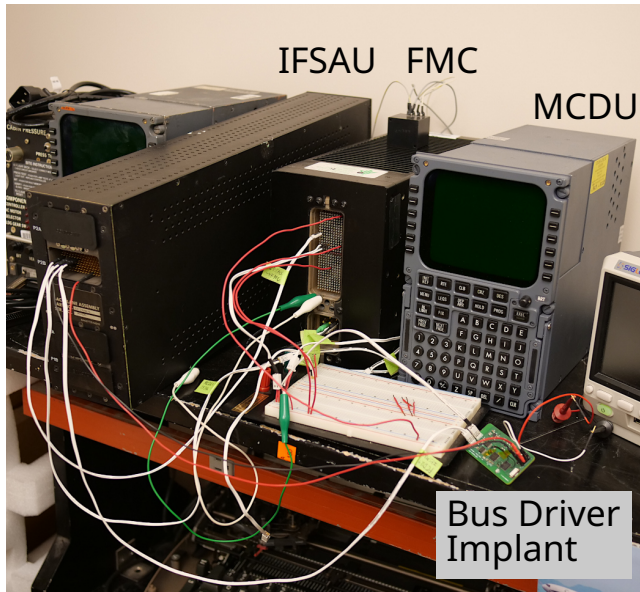


Figure 7: Our 737 FMC/MCDU test setup with the main components labeled. The implant shown here is larger than the one in Figure 10 because this version includes additional space between the components for easier debugging.

removed from in-service aircraft. While there are newer versions of some of these parts, they are completely interchangeable and we have verified that their behavior is the same with respect to our attack. The FMC we tested is a GE 2907A4 that previously served on a Southwest Airlines 737 and our MCDU is a GE 2577F1 from a Batavia Air 737. We also acquired a genuine IFSAU and an aviation connector with the same pinout and form-factor as the OMC.

We duplicated the exact wiring of the OMC between the FMC and MCDU units, including the Integrated Flight Systems Accessory Unit (IFSAU) which houses the OMC-controllable relay switching the key-press input to the FMC (Figure 3). We connect all devices to a shared ground bus, as they are in the aircraft (Figure 7). The lab-bench testbed we created is as close as possible to the way that aircraft manufacturers test their avionics systems [12].

Finally, after our own testing described below, Boeing generously and graciously allowed us to test a version of our prototype hardware in *their* 737 testbed. We were able to validate that our Bus Driver-based attack worked in their environment as well and were able to mediate FMC and MCDU communications as we expected.

5.2 Bus Driver and the MCDU protocol

To avoid detection, the attacker must be able to override the characters that the FMC sends via the MCDU’s character-based protocol. Each 32-bit ARINC 429 message encodes one character on the MCDU’s screen. However, updates are



Figure 8: MCDU “FAIL” light illuminates if the Bus Driver does not follow the timing of ARINC 429 32-bit messages.

run-length encoded so a single character can be repeated using only one additional message. For instance, when a line on the screen has N blank spaces in a row, the FMC simply sends a single 32-bit space character that is run-length encoded with a length of N . The MCDU screen is drawn an entire screen at a time, starting from the top left of the screen to the bottom right, with a sequence of three magic messages that indicate a new page. The screen is refreshed regularly in this manner. The messages have specific timing that must be met. If not, a “FAIL” light will appear on the MCDU (Figure 8), the MCDU will return to the main menu, and the FMC will be briefly unavailable on the MCDU until the proper pacing returns. Such an error message could easily raise suspicions as it does not occur during normal flight.

Thus, there are key practical challenges in using the Bus Driver AITM capability to modify characters sent to the MCDU screen. Since the MCDU will show a FAIL message if the constant text message stream is interrupted, the implant must constantly receive messages to synchronize its timing so that its overridden messages exactly overlap the FMC’s messages. The FMC sends all the messages that make up a screen update in rapid succession with a gap of $40\mu\text{s}$ between messages. When the implant modifies one of those messages, it needs to send its message within approximately $10\mu\text{s}$ of the ideal time. If the message is too early or late, the gap before or after the message will be too short and the MCDU will illuminate the FAIL light. Further, a subtle problem can arise when overriding a repeated character, which then requires adding more 32-bit messages than the FMC originally sent to write the screen—leading the Bus Driver and the FMC to lose message synchronization. In such scenarios, the implant must continually operate in overwrite mode and correctly model the contents of the screen as needed.

5.3 End-to-end Attack

In spite of these practical challenges, we have still been able to prototype a full end-to-end attack via the OMC, as follows:

1. Record the current screen update the FMC is sending to the MCDU
2. Freeze the screen content by repeatedly sending (i.e., overriding) the same page to the MCDU.
3. Trigger the relay to take input away from MCDU and inject a series of malicious button presses to the FMC (e.g., navigating to sensitive pages, updating their state, and navigating back to the original page).

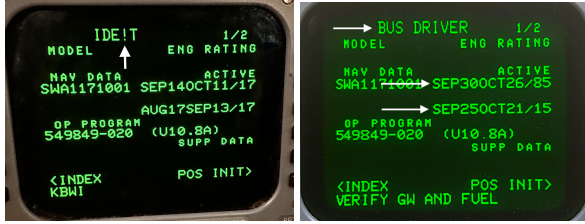


Figure 9: Examples of selective text modification (left) and full screen override (right) of the FMC's default IDENT page.

- Record and store the original state overwritten by the injected button presses, so this prior state may be “played back” to hide the changes to the FMC.

Once these malicious changes to the configuration of the FMC are complete, the attack only requires hiding the changes from any checks the pilot may do on the MCDU to see what is going wrong. Here there are two options depending on the nature of the state modification:

Selective text modification. If the changed parameter or value is the same length as the original value (e.g., KBWI way-point changes to KDCA), the implant can hide these changes using “trigger patterns.” In this scenario, the implant allows MCDU messages to pass through unchanged *unless* they contain the particular trigger term (e.g., a field name label). When the implant witnesses such a trigger it enables its output and overwrites the subsequent characters with the appropriate (i.e., incorrect) values it overwrote during the injection phase of the attack. As a simple example, the left-hand side of (Figure 9), illustrates the result when the implant detects the sequence “IDE” (from the IDENT page) and replaces the next character (“N”) with an exclamation point.

Full text modification. If the changed parameter is a different length (e.g., the speed 10 is replaced by 100 and that changes the number of repeated characters (e.g., spaces before the numbers)), the implant must fully override the MCDU screen for the rest of the flight. The challenge is maintaining interactivity to the pilots to avoid detection. To achieve this, the implant can use its listen-while-overriding current sensing circuit to capture the characters the FMC is sending, modify them as needed, and then send the complete pages to the MCDU while it is continually overriding. The right-hand side of Figure 9 shows such a modification (including the string “BUS DRIVER” and modified NAV DATA dates).

5.4 “Productizing” the implant

Practical deployment of such an implant must overcome two other practical concerns: power and stealth. Conveniently the OMC connector provides constant power (both on the



Figure 10: Implant with integral Wi-Fi connectivity in form factor that can be hidden almost entirely inside the body of the OMC.

ground and in the air) which is sufficient to power the implant. Stealth is a more significant engineering concern as any other personnel entering the E&E bay might notice unfamiliar equipment connecting to aircraft systems. To this end we have designed and fabricated a custom multi-layer circuit board to miniaturize our design. As shown in Figure 10, this version of our implant fits almost completely inside the OMC housing. Moreover, since the OMC is normally covered by a dust cap, our prototype can be hidden completely below this cap and can wait there until activated.

Finally, while such an implant can be designed to activate in response to triggers in the MCDU (e.g., entering a flight plan), the current time, or based on atmospheric sensors (e.g., pressure), we have also incorporated Wi-Fi into the hardware design. The premise is that such an implant could connect to the passenger’s in-flight Wi-Fi and then relay, via the Internet, to a remote controller. While we believe it is likely that the forward cabin Wi-Fi signal would be able to penetrate the floor to the avionics bay, we have not been able to test this empirically.

5.5 Attack Scenarios

The 737’s OMC allows an implant to control all of the functionality of the FMC that is accessible to pilots from the MCDU. In reviewing these capabilities—both before takeoff and during flight—we identified three that we found the most concerning for flight safety and that have been previously implicated in accidents and near-accidents [14].

Execute flight route change. Perhaps the most significant threat posed by maliciously controlling the FMC is re-routing the plane in flight. It is common for 737 pilots to activate the autopilot and autothrottle during cruise. In this mode, its Flight Control Computer (FCC) directly manages thrust and control surfaces to navigate to the flight plan waypoints provided by the FMC. An implant could insert a new waypoint by surreptitiously navigating to the LEGS page on the MCDU

and adding a direct route to a new waypoint at the top. The FMC is designed to make pilots confirm such route changes which then take effect immediately (i.e., the autopilot will immediately begin to steer toward the new waypoint). Thus, after a route change the EXECUTE button on the MCDU will illuminate and the pilot must press this button to confirm. However, since both the EXEC button and its illumination are controlled via ARINC 429 messages on the same bus, this safeguard can be easily bypassed. To be clear, the implant is able to “press” the EXEC button on the pilot’s behalf (i.e., with no human involvement)—**causing the plane to immediately turn towards the new waypoint**. Moreover, should an attacker desire to act covertly, the illumination of the button can be suppressed and the updated contents of FMC-related MCDU pages can be overwritten, thus hiding the fact that a new route has been loaded and is being followed.

Alter Weight and Balance. Another serious threat is that an attacker can modify the FMC’s input of the “Zero Fuel Weight” (ZFW) of the aircraft, accounting for passenger and baggage weight (and related parameters). The per-flight value is typically provided directly by the airline (e.g., via ACARS message) or calculated indirectly via an external iPad. The ZFW is critical to the FMC’s calculations of how the plane will operate during many safety-critical aspects of flight, including during takeoff (e.g., V-speeds) and while in-flight (e.g., fuel burn rate and climb and descent rates, and maximum cruise altitude reachable). However, once the pilot has set it, **the implant can silently overwrite the ZFW**. Of primary concern is that the FMC might compute the takeoff speed and thrust settings needed based on incorrect aircraft weight, and leave the plane at risk of being unable to safely take off.

Alter assumed temperature. A final but related threat is that takeoff thrust settings are calculated based on the assumed outside temperature. Pilots often increase the assumed temperature to have the FMC leave more generous margins for takeoff performance settings required. However, **the implant can silently lower that margin by lowering the assumed temperature**, creating another situation where there may be insufficient thrust to complete takeoff.

6 Defenses

Having outlined these potential issues, we now consider defenses against the class of attacks we have described thus far. In particular, we focus on technical interventions beyond merely improving physical access controls around aircraft (i.e., defenses that are effective in spite of the adversary obtaining temporary physical access). We divide these approaches into three rough categories: those that physically or electrically prevent Bus Driver attacks, those that detect

that a Bus Driver attack is occurring via hardware measuring changes to the electrical properties of the bus, and those that detect an attack via software changes to higher levels of the protocol stack.

Each defense has drawbacks in the form of removing functionality, requiring new hardware or software, or introducing new challenges such as cryptographic key management.¹⁶

In general, the defenses which merely detect, but do not prevent, attacks come with an added challenge: What do you do once an attack has been detected? For the specific vulnerability we identified in the open maintenance connector connected to the captain’s FMC and MCDU, if a Bus Driver attack were detected, the flight crew might switch to using the first officer’s FMC (see Section 3.1) and revalidate its contents, but this approach may not generalize to other scenarios.

6.1 Prevention

An ideal solution would be to remove the underlying capability to access sensitive buses via open connectors and to prevent the synthesis of AITM capabilities even when such buses are accessible. We consider both approaches in turn.

Protecting the connector. The attack we have described is enabled by the particular capabilities of the OMC, and if these were foreclosed, then our particular attack would not be possible.

Perhaps the simplest form of such a defense would be to remove the connector entirely or fill it with epoxy, physically preventing a Bus Driver attack using the connector. However, such a choice would also limit any utility it might have for maintenance purposes. Moreover, this would address just a single connector and there are a variety of maintenance connectors that might enable other, similar attacks.

An alternative approach is to protect such connectors electrically. A key component of the success of the Bus Driver attack on ARINC 429 is that legitimate transmitters include 37.5 Ω resistors but the open connector provides direct bus access. This direct connection enables an implant to directly drive the bus to the target voltage while the legitimate transmitter is limited by the resistors. Thus, if the resistors were incorporated into the connector itself rather than in the transmitter, the Bus Driver attack would fail. In this scenario, both the implant and legitimate transmitter would be drawing the same current through the resistors and the voltage on the bus would go to 0 V.

An even stronger form of this defense, as we discuss below, is provided by the MIL-STD-1553 bus, which is transformer-coupled rather than resistor-coupled. In this bus the transformers are physically housed in the bus couplers and any implant connected using a standard coupler is thus prevented from directly driving the bus.

¹⁶We are unaware of any such defenses in use, or even in development, for existing 737s or similar commercial transport aircraft.

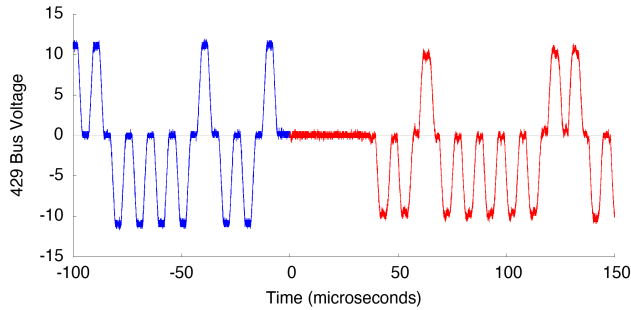


Figure 11: The ARINC 429 bus waveform during a legitimate transmission (blue) is slightly different at the peaks and troughs of the signal during the Bus Driver attack (red).

Galvanic isolation of devices on the bus. As described previously, each ARINC 429 device connects to the bus through a pair of resistors. The Bus Driver attack succeeds because the implant can be connected directly to the bus without resistors and thus override a legitimate transmitter.

One approach to prevent Bus Driver attacks is to galvanically isolate devices from the bus, that is, prevent current from flowing from any device to the bus. In order to communicate on the bus, devices need to use some other coupling mechanism such as a transformer or an opto-isolator. Using such a bus architecture prevents a Bus Driver implant from overriding legitimate messages by ensuring that all devices on the bus—including the implant—have exactly the same ability to manipulate signals on the bus.

MIL-STD-1553 is an example of a transformer-coupled bus used in military aviation. Appendix D describes MIL-STD-1553 in more detail and shows that its design is robust enough to withstand even stronger forms of the Bus Driver attack we have described thus far.

The downside to switching to a new bus architecture, such as MIL-STD-1553, for commercial aviation is clear: Changing bus architectures requires changing the communication hardware for every avionics system currently using ARINC 429.

6.2 Detection via electrical measurements

Today’s ARINC 429 chipsets do not integrate voltage-based detection of faults on ARINC 429 buses. We describe a defense based on the anomalous voltages the Bus Driver will introduce into the ARINC 429 bus. We describe how these differences are visible both at the ARINC 429 receiver and transmitter.

Detection via voltage at the receiver. We evaluate the feasibility of monitoring voltage of the bus to detect the Bus Driver overriding ARINC 429 messages. Previous work demonstrated it is possible to fingerprint ARINC 429 transmitters

based on their voltages on a bus [17]. Since the Bus Driver transmits at the same time as the legitimate transmitter, the modified ARINC 429 waveform would likely be detected by voltage-based intrusion detection systems. Figure 11 shows the waveform of a legitimate transmission (blue) followed by a overridden transmission (red). The Bus Driver overrides the legitimate signal entirely, and although it produces a nearly identical waveform as the legitimate transmitter, there are slight differences. The Bus Driver’s waveform includes more noise, especially at the peaks and troughs—likely from the Bus Driver’s high-gain amplifier—and the voltage on the bus is not exactly the same as the legitimate transmitter. Since they are not identical, it would be possible to create a defense against this attack by sampling the full analog waveform of the bus and finding incongruities using digital signal processing [17].

Detection via voltage at the transmitter. When an implant is actively driving the bus, the voltage on the bus is different from what the legitimate transmitting device wants. One way to detect this situation is to have an additional receiver connected to the bus near the transmitter. This receiver would measure the actual voltage on the bus. If the voltage the receiver senses does not match what the transmitter is trying to send, there is a problem.

Some other buses use a similar feature. For example, CAN bus transmitters measure the voltage on the bus and compare it to the bits being transmitted. This functionality is crucial for CAN bus arbitration. Specifically, if one transmitter transmits a logical 0 at the same time a second transmitter transmits a logical 1, both transmitters will measure a 0 on the bus and the second transmitter will stop transmitting. This ensures that if two transmitters start sending a CAN frame at the same time, the frame with the lowest numerical ID value wins arbitration. When sending the data portion of the CAN frame, an incorrect bus voltage causes the transmitter to stop sending and to signal an error.

One approach to add a voltage measuring defense to an ARINC 429 network is to redesign the network so that each LRU has a dedicated receiver for each transmitter. Software on the LRU would need to receive each message it transmits and compare with the original sent message and signal an error on a mismatch. For example, ARINC 702A [1] specifies that a Flight Management Computer (FMC) has 19 ARINC 429 transmitters and 38 receivers, including 4 spare transmitters and 3 spare receivers. FMCs would need to add at least a dozen additional receivers to have a receiver per transmitter.

Detection via current. In addition to detecting Bus Driver attacks by measuring voltage at the receiver or transmitter, a Bus Driver attack can be detected by measuring the current flowing through the bus.

As described in Section 4.3, the 429 implant drives about 267 mA of current through the bus when it is in conflict with

another transmitter. That is significantly more current than during normal operation. An added device near each transmitter could measure the current flowing through the bus and detect unusually large amounts of current. This could be integrated into the transmitter or in a separate unit, as long as it is between the transmitter and the implant.

This is perhaps more practical than measuring the voltage because it could be added to existing buses without replacing each individual LRU.

6.3 Detection via software

A major drawback of both measuring the voltage and measuring the current on the bus to detect a Bus Driver attack is it requires additional hardware. Software-only defenses may be easier to deploy but would require significant changes to the ARINC 429 protocol stack.

The fundamental danger of a Bus Driver attack is that the targeted device will act on the illegitimate messages sent by the implant. One standard way to ensure that received messages are sent by a legitimate transmitter is to use cryptography. Specifically, one can change the higher-level protocols used on top of ARINC 429 to attach a message authentication code (MAC) to each message.

There are two key downsides to introducing MACs in the higher-level protocols. First, most ARINC 429 messages consist of just a single 32-bit data word. Adding an authentication tag would necessitate sending a second word,¹⁷ significantly increasing the amount of traffic sent on the bus. Second, as with most problems where adding cryptography is the solution, using a MAC introduces a key management problem. MAC keys need to be exchanged between communicating LRUs and securely rekeyed when an LRU is replaced or updated.

7 Related work

In general, we are unaware of any work that empirically evaluates cybersecurity questions in the context of real transport aircraft such as the 737 (likely due to the challenges in obtaining such artifacts for testing). However, our work benefits from the legacy of prior efforts focused on the security of industrial communications. Below, we summarize some of the most closely related work concerning injecting, blocking, and modifying messages on aviation and automotive buses—both attacks and defenses.

7.1 Attacks

Prior attack work typically involves either attaching a new device to the bus or leveraging an existing, compromised,

device on the bus, in order to inject and manipulate traffic. These efforts are broadly similar across analyses of CAN, ARINC 429 and MIL-STD-1553, but the underlying nature of the physical bus protocol places different limitations on the attacker.

CAN bus attacks. The CAN bus has attracted significant research interest because it is widely used in almost all modern automobiles (and hence readily available for testing) and is often implicated in safety-critical communications. Moreover, as CAN is a multi-master bus, with no built-in authentication features, any device on the bus can send any message—making CAN naturally vulnerable to injection attacks (e.g., see Checkoway et al. [6]). As well, similar to our own work, many of the threat models for automotive CAN exploit the wide availability of an accessible maintenance connector with bus access (OBD-II in automobiles).

A number of researchers have also demonstrated the ability to *disable* the CAN bus by inducing errors during legitimate transmissions—i.e., message suppression [7, 16, 31]. However, a true AITM capability on CAN is challenging as its physical layer uses open-drain circuits. Thus, while any number of nodes can concurrently assert the dominant state, the recessive state can only occur when no node is active [23]—i.e., a CAN implant can change a ‘1’ bit to ‘0’, but not the reverse.

Perhaps the closest approximation to an AITM capability in CAN is demonstrated by Hoppe et al. [21] who showed, in a lab environment, that an attacker could implicitly “override” legitimate commands (under an implicit assumption of statelessness), by detecting their transmission and immediately sending *another* message with a different command. In their environment, the device processing the commands would effectively ignore the legitimate command and follow the malicious command because it arrived later. By contrast, in our work we do not need to rely on the stateless nature of higher-layer application semantics, but are able to actively suppress, read and modify legitimate messages at the physical layer of the ARINC 429 protocol.

429 bus attacks. As with CAN, ARINC 429—widely used in aviation—has no inherent security features and thus no inherent protection from message injection. However, unlike CAN, ARINC 429 is a single master bus and, as a consequence, connecting multiple legitimate 429 transmitters to a single bus “violates the electrical properties of the system” and “[t]herefore, an adversary cannot simply add a transmitter [...] to the bus” [17]. Adding a second transmitter “reduces line impedance, prohibiting receivers from reading data successfully” in a test setup [4]. Thus, prior work has primarily embraced a threat model whereby legitimate transmitters or receivers are replaced and/or compromised [4, 5, 17]. Our work overcomes this limitation using custom hardware that does not violate the electrical properties of the bus. This has two

¹⁷The 32-bit data words are divided into fields depending on the message. Some messages use just a single, small field. For those cases, it may be possible to include a MAC tag in unused bits in the data word.

implications. First, since our attack does not require replacing the legitimate transmitter, it opens up the “connector-based” attack surface. Second, because the Bus Driver implant can measure the bus current to read what message the legitimate transmitter is sending while simultaneously overriding that message, it is possible to implement a true AITM capability.

MIL-STD-1553 attacks. Finally, military avionics frequently make use of the MIL-STD-1553 bus, which depends on a bus controller to orchestrate bidirectional communication between a set of attached remote terminals. De Santo et al. [13] investigate threats against this bus—again, exploiting the lack of any inherent security features in the protocol. They show that a malicious device on the bus can both “jam” the bus (i.e., via traditional capacity-consumption denial-of-service methods) or, by waiting for idle periods, can inject spoofed messages as well. Note that the requirement to successfully anticipate idle periods is precisely because overriding a legitimate transmission is extremely difficult in MIL-STD-1553 (as we document in Appendix D).

7.2 Defenses

Prior defense research in this space is generally focused on detecting anomalies. The defender measures electrical or timing properties of legitimate communications on a bus and then, during operation, identifies deviations from this learned electrical or timing profile due to an attack.

Voltage analysis defenses. Several researchers, both for CAN [24, 34] and ARINC 429 [17], have proposed measuring the voltage over time on a bus and analyzing the readings to detect when a device has been added, removed, or substituted. This approach would be able to detect the type of attack described in this work. Even from casual inspection with an oscilloscope, a message modified by an implant looks different from an unmodified message.

The main problem with this approach is that it requires an analog-to-digital converter running at several million samples per second, and non-trivial computing resources to process the samples and detect anomalies. Practical deployments also must address natural variances, e.g., on vehicles, shifts in temperature or primary battery voltage can cause voltage shifts which fingerprinting must account for [8]. Models also need to be re-calibrated to prevent false positives when a device on the bus is replaced for benign reasons.

Timing defenses. Another common approach for detecting bus traffic anomalies is the timing of messages. For example, a wide array of proposed CAN bus intrusion detection systems [26] assume periodic message scheduling and are able to detect deviations from a learned inter-arrival pattern. G  n  reux et al. [19] propose an intrusion detection system

for MIL-STD-1553 using a range of timing features, including the amount of time between messages and the precise delay before a device responds to a command from the bus controller. Similar features appear in later work by Babir et al. [4] focused on detecting transmission anomalies on ARINC 429 buses.

It seems unlikely that such defenses would be effective in detecting the attacks described in this paper since the Bus Driver attack changes the content of messages, and thus, the overall message timing stays the same. However, if the bus were physically large and the implant was far from other devices, it is conceivable that very fine-grained measurements could detect differences in propagation delay.

8 Conclusion

This paper is driven by three goals. The first is to help both the security and aviation communities reconsider a seemingly implicit assumption that aviation threats predicated on physical access do not deserve significant attention. We believe we have made a strong case that time-limited physical access (e.g., 60 seconds) represents a realistic goal for a motivated attacker and that the consequences of even such short access can be significant (and hence are worthy of attention). Second, we also hope to highlight our observation that an Attacker In the Middle (AITM) capability on a bus is possible without physical mediation (i.e., without “being in the middle” *per se*). We have argued and demonstrated that the ARINC 429 bus (and buses with similar electrical properties) are vulnerable to Bus Driver attacks, whereby an adversary with bus access can amplify that capability to mediate all other communications on the bus. This is both an important realization for threat modeling fielded aircraft, and also a design consideration for future bus choices and designs. Third, we have identified a specific and detailed scenario, and tested it empirically, by which the combination of these two previous factors exposes a new threat on the Boeing 737 aircraft. An attacker with short-term access to the plane’s E&E bay can insert a miniature implant into an existing connector, which in turn is able to block and modify messages in both directions between the Flight Management Computer (FMC) and the Multi-function Control and Display Unit (MCDU)—a channel with significant flight safety implications. We have demonstrated this capability empirically—showing that an attacker can surreptitiously change the state of the plane’s FMC and then subsequently conceal these changes from the pilot. This proof-of-concept should illustrate that this is not merely a theoretical issue, but one that can be engineered practically today. Indeed, while we focused singularly on the OMC in our work, we are aware of many such maintenance connectors in the E&E bay whose attack surfaces should also be rigorously evaluated in the light of our findings.

Finally, we note that complex industrial systems such as transport aircraft are frequently seen as outside the practical

scope of all but the most well-funded and resourced attackers (e.g., state actors). We hope our results, primarily the product of a single graduate student with modest financial resources, helps people reconsider this assumption as well.

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A Ethical Considerations

This paper investigates the feasibility of attacks on the Boeing 737 that amplify temporary physical access into a full-fledged attacker-in-the-middle capability by both manipulating the ARINC 429 data bus to override signals, exploiting the design of particular maintenance connectors on the Boeing 737 and careful reverse engineering of the message structure and operating behavior of the standard Flight Management Computer (FMC) and Multi-function Control and Display Unit (MCDU) used in that airframe. The potential harms from this work arise from educating attackers interested in targeting such environments, balanced against the benefits from understanding these risks and helping to mitigate similar threats in the future. In general, we believe this balance favors publication because our work is speculative in nature (i.e., we are unaware of any active attackers in the wild who this work would enable) and because it highlights a class of vulnerabilities that have not

traditionally been considered in scope and we believe require attention.

Moreover, we have provided significant advance disclosure. In particular, we disclosed to Boeing in April of 2020, with multiple follow-up meetings and discussions over the next several years (in-person meetings, where we made our hardware and personnel directly available, were delayed by the COVID-19 pandemic until 2023). Boeing, in turn, disclosed to the Aviation Cyber Initiative (ACI), an aviation cybersecurity forum jointly chaired by the FAA, DHS and DoD. We also disclosed informally at a closed aviation security meeting which included representatives of the FAA, DHS, the major US passenger airlines, large aviation equipment manufacturers (e.g., Honeywell, Collins, GE) as well as Boeing and Airbus. Finally, we also disclosed in the course of extended conversations with a pilot and security representative of a major US passenger airline who helped us validate our understandings of flight operations and their relationship to the underlying vulnerability.

However, transport aviation can be a challenging domain to mitigate—it is global, federated (there are many distinct owners of Boeing 737 airframes) and heavily regulated. Any changes to hardware and software can be expensive and time-consuming to deploy and must be balanced against any potential new risks they create. As of this writing, we are unclear on the precise extent to which Boeing has been able to mitigate this vulnerability or similar vulnerabilities they have discovered in their investigations (see their statement in Appendix C). As a result, we have chosen to omit certain details not critical to understanding the underlying issues, but important to mounting the concrete attack in practice (e.g., the name and location of the vulnerable access port, the schematics of its pinout, and details of the proprietary MCDU protocol). Boeing agrees with this decision on our part.

B Open Science

This is a situation where ethical interests and open science interests pull in different directions. The attacks we have described have clear flight safety implications and, as we have said, we are unable to independently establish that Boeing has eliminated this issue in fielded aircraft. These factors weigh strongly against sharing our artifact (a piece of hardware) and some of its details with the public. However, we observe that the most generalizable technical idea behind our attack is easy to replicate based entirely on the text of our paper (e.g., overriding the ARINC 429 bus) without needing any more detail than we describe herein. However, replicating the specific detailed attack described in this paper (i.e., against the MCDU/FMC/IFSAU combination) is unlikely to be feasible for many parties as it will require access to a Boeing 737 (or alternatively, at least the three key LRUs, the particular maintenance connector and knowledge of the appropriate wiring between them). Based on this analysis, and after consultation

with the PC and REC chairs, we have chosen to limit our artifact release to the paper itself (notably the circuit description figure therein).

C Boeing Disclosure and Response

As we explained earlier, we initially disclosed our findings to Boeing in April of 2020, followed by multiple briefings to company representatives and culminating with successful hardware tests on Boeing’s own 737 testbed in December of 2023. On the basis of those outcomes, we understand Boeing undertook further efforts to investigate and address this concern. As per their practice, the precise nature of their findings, or any mitigations that might have resulted, is not public, but have they provided the following statement for inclusion here:

We are grateful for UCSD’s responsible disclosure and collaboration on their 737 ARINC 429 Flight Management Computer (FMC) and Multi-purpose Control Display Unit (MCDU) research. Their professionalism throughout our engagements is commendable.

Upon review of the researchers’ work, and controlled demonstrations with them in our laboratory environment, Boeing completed a thorough evaluation and assessment of the FMC design, systems installation, and interfaces. Our technical experts are confident that the layers of protection in place on the airplane, including within the system design and the operating environment, provide sufficient mitigation to significantly limit the feasibility and risk of real-world attacks.

Safety is the foundation of everything we do, and we take threats to our products seriously. We encourage researchers to responsibly reach out to Boeing with concerns regarding product security.

D MIL-STD-1553

We have demonstrated that ARINC 429’s design as a resilient industrial bus made it particularly vulnerable to Bus Driver attacks. There is another bus standard, MIL-STD-1553, which is used in military aircraft [35] and some weapon systems (e.g., the Javelin missile).

We briefly describe the relevant differences between ARINC 429 and MIL-STD-1553 and show through testing that even in a best-case scenario, MIL-STD-1553 withstands Bus Driver attacks.

The key difference between MIL-STD-1553 and ARINC 429 is that MIL-STD-1553 incorporates coupling transformers between the main bus wiring and the devices on the bus. These transformers provide galvanic isolation—an air gap

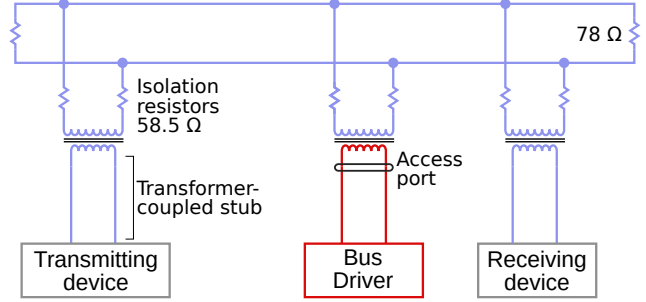


Figure 12: A MIL-STD-1553 bus’s transformer-coupled stubs offer protection against a Bus Driver implant.

that makes it infeasible for current to directly flow between the bus and the attached devices. The coupling transformers must be located near the main bus wiring and are part of the couplers themselves.

With transformer coupling, the amplifier-based attack circuit we described for ARINC 429 would not work. The problem is the amplifier’s overriding would only affect signals on the same side of the transformer that the attacker is on. The other parts of the bus will still have the legitimate transmission (Figure 12).

In theory, having multiple connections to the same bus may allow a Bus Driver attack (Figure 13). Specifically, one bus connection senses the state of the bus which is inverted, amplified, and injected back onto the bus using the second connection. If this cancellation signal were instead connected directly back to the sampling input (i.e., injected onto the same bus connection that was used to listen to the bus), the slight phase difference between the original and cancellation signal would be fed back into the cancellation circuit and amplified over and over [30]. Using two connections avoids this.¹⁸

Since some MIL-STD-1553 bus coupling modules include multiple independent connection ports on a single device [9], finding two open connections may be feasible. As such, we evaluate the efficacy of a Bus Driver attack in this configuration with an implant connected to the bus in two locations.

Specifically, we use an STMicroelectronics STM32F446 microcontroller to poll one of the MIL-STD-1553 connections to detect and read legitimate transmissions. Once a transmission is detected, the microcontroller uses bit-banging¹⁹ to send a cancellation signal on the other MIL-STD-1553 connection.

One challenge is that the MIL-STD-1553 standard requires receivers to tolerate significant attenuation: an attack has to

¹⁸For an intuitive analogy, consider noise canceling headphones. The microphone sensing noise is separated from the speaker that outputs the cancellation signal into the ear such that the cancellation waveform is not picked up by the microphone.

¹⁹Experiments with the quad SPI peripheral included with the STM32F446 produced the correct outputs but with unpredictable delays.

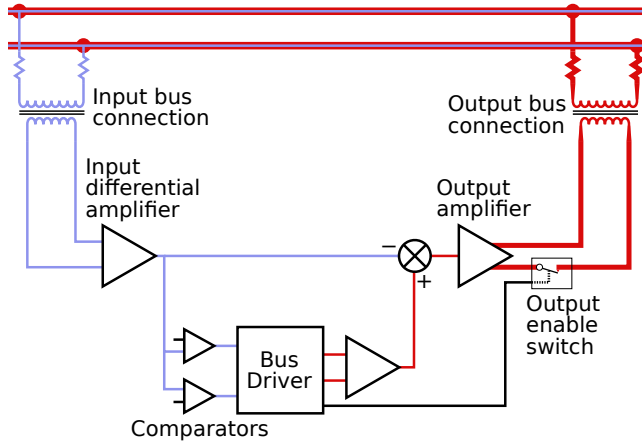


Figure 13: A block diagram of a MIL-STD-1553 Bus Driver attack which requires two connections to the bus.

cancel the legitimate signal from up to 13 V down to under 0.6 V [35].

Our experiments show that although a Bus Driver implant can significantly attenuate the legitimate signal, it is unable to drop the voltage enough to alter or cancel the signal. Figure 14 shows a legitimate MIL-STD-1553 waveform compared to the waveform when the Bus Driver implant tries to cancel the transmission. Specifically, it does not cancel the beginning and end of each bit as there is a delay in the cancellation circuit's inverting amplifier that is proportional to its bandwidth. During the transmission of each bit, the implant decreases the amplitude of the bit substantially, but not below the 0.6 V threshold.

We tried several high-gain amplifiers, including amplifiers designed specifically for this type of application, namely long-distance telecommunication systems (e.g., VDSL). We find that the higher the output voltage, the more likely it was that the cancellation circuit would end up in an oscillation feedback loop, a classic problem with high-gain feedback circuits.

Overall, galvanic isolation from transformers significantly mitigates Bus Driver attacks. It requires an attacker find multiple unused bus connections and limits the range of voltages which can be canceled effectively.

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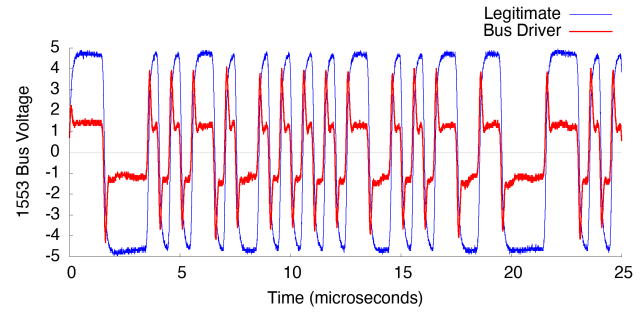


Figure 14: A legitimate MIL-STD-1553 waveform compared to the Bus Driver's attempt to cancel the waveform.

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